

LEARNING TASK 3

Course ID: Probability & Statistics

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Discrete Random Variables

Given a random experiment E , for example eliminating a target in a maximum of five attempts, with a corresponding sample space S

$$S = \{h, mh, mmh, mmmh, mmmmh\},$$

where m denotes a miss and h denotes a hit.

Let X be the number of the attempt at which the target is ruled out. The range of X is $\{1, 2, 3, 4, 5\}$, hence X is a discrete random variable.

Probability Mass Functions & Cumulative Distribution Functions

Definition 3.1: PMF & CDF

The **probability mass function** (pmf) is defined as $f(x) = P(X = x)$.

The **cumulative distribution function** (cdf) is defined as $F(x) = P(X \leq x)$.

Example 3.2

Ruling out a target in a maximum of five attempts. Suppose that the probability of hitting at attempt i , given that the previous $i - 1$ attempts were misses, is equal to $\frac{i}{5}$. Determine the probability mass function $f(x)$ and the cumulative distribution function $F(x)$ of X . Draw the graphs of $f(x)$ and $F(x)$.

Solution

The probability mass function values are calculated as follows:

$$P(X = 1) = P(h_1) = \frac{1}{5} = 0.2000$$

$$P(X = 2) = P(m_1 h_2) = P(m_1) \cdot P(h_2 | m_1) = \left(1 - \frac{1}{5}\right) \cdot \frac{2}{5} = 0.3200$$

$$P(X = 3) = P(m_1 m_2 h_3) = P(m_1) \cdot P(m_2 | m_1) \cdot P(h_3 | m_1 m_2) = \left(1 - \frac{1}{5}\right) \cdot \left(1 - \frac{2}{5}\right) \cdot \frac{3}{5} = 0.2880$$

$$P(X = 4) = \dots = 0.1536$$

$$P(X = 5) = \dots = 0.0384$$

The probability mass function for X can also be described as a table:

x	1	2	3	4	5
$f(x)$	0.2000	0.3200	0.2880	0.1536	0.0384

Figure 1: Probability mass function of the discrete random variable X .

The cumulative distribution function $F(x)$ is evaluated at piecewise intervals:

$$F(x) = \begin{cases} 0.0000 & x < 1 \\ 0.2000 & 1 \leq x < 2 \\ 0.5200 & 2 \leq x < 3 \\ 0.8080 & 3 \leq x < 4 \\ 0.9616 & 4 \leq x < 5 \\ 1.0000 & x \geq 5 \end{cases}$$

Notice that the cumulative distribution function is a step function, in which vertical steps occur at each of the possible outcomes of X , and the height of the jump at outcome x is equal to $P(X = x)$.

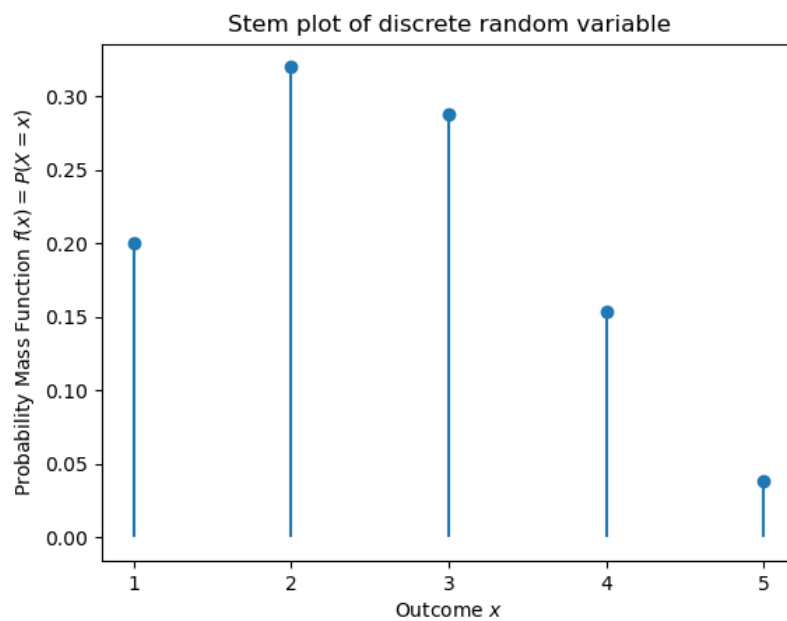


Figure 2: Graph of the probability mass function of X .

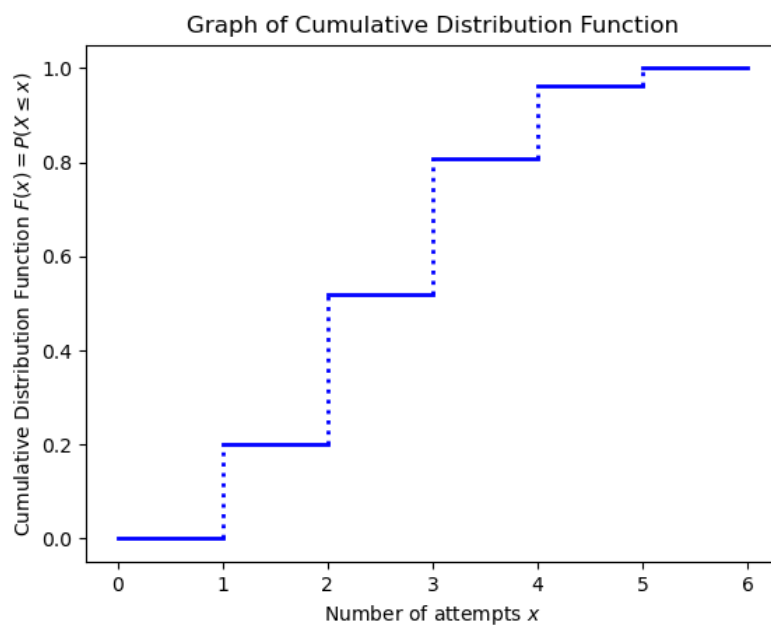


Figure 3: Graph of the cumulative distribution function of X .

General Properties

- $f(x) \geq 0$ for all x , and $\sum f(x) = 1$.
- $F(x)$ is a non-decreasing step function.

Mean and Variance

Definition 3.3: Expected value and variance of a random variable

The *expected value* of a random variable X is defined as

$$E(X) = \sum_{x \in S} x \cdot f(x),$$

where $\sum_{x \in S}$ denotes the sum over all possible outcomes x in the sample space S . Similarly, the *variance* of a discrete random variable X is defined as

$$\sigma^2 = V(X) = \sum_{x \in S} (x - E(X))^2 \cdot f(x).$$

As a result of this definition, one can also define the expected value of a *function* of the random variable X :

$$E(g(X)) = \sum_{x \in S} g(x) \cdot f(x).$$

An alternative formula for the variance $V(X)$ is as follows:

$$V(X) = E(X^2) - [E(X)]^2.$$

Example 3.4

Using data from the target engagement distribution:

$$E(X) = 1 \cdot (0.2) + 2 \cdot (0.32) + 3 \cdot (0.288) + 4 \cdot (0.1536) + 5 \cdot (0.0384) = 2.5104$$

$$E(X^2) = 1^2 \cdot (0.2) + 2^2 \cdot (0.32) + 3^2 \cdot (0.288) + 4^2 \cdot (0.1536) + 5^2 \cdot (0.0384) = 7.4896$$

$$V(X) = E(X^2) - [E(X)]^2 = 7.4896 - (2.5104)^2 \approx 1.1875$$

The standard deviation is $\sigma = \sqrt{V(X)} \approx 1.0897$.

Linear Transformation Properties For constants a and b :

$$E(aX + b) = aE(X) + b$$

$$V(aX + b) = a^2V(X)$$

Cost Function Example Let total cost $C = 1000X + 2000$, with €2.000 fixed costs and €1.000 variable cost per attempt:

$$E(C) = 1000E(X) + 2000 = 1000(2.5104) + 2000 = \text{€}4.510,40$$

$$V(C) = 1000^2V(X) = 1000^2(1.18749184) = 1.187.491,84$$

$$\sigma_C = \sqrt{V(C)} = \text{€}1.089,72$$

Discrete Uniform Distribution

A discrete random variable X has a discrete *uniform* distribution over n values $x_1, x_2, \dots, x_n$ if:

$$P(X = x_i) = \frac{1}{n} \quad (i = 1, 2, \dots, n)$$

If the range consists of n consecutive integers $a, a + 1, \dots, b = a + n - 1$, the mean and variance are:

$$E(X) = \frac{a + b}{2}$$
$$V(X) = \frac{(b - a + 1)^2 - 1}{12}$$

Example 3.5: Fair die

For a fair six-sided die, $X \in \{1, 2, 3, 4, 5, 6\}$ with $a = 1, b = 6, n = 6$:

$$E(X) = \frac{1 + 6}{2} = 3.5$$
$$V(X) = \frac{6^2 - 1}{12} = \frac{35}{12} \approx 2.9167.$$

Example 3.6: Coating thickness

Thickness measurements $X \in \{2.1, 2.2, 2.3, 2.4, 2.5\}$. Let $X = 0.1Y$, where Y is uniformly distributed over integers $a = 21$ to $b = 25$:

$$E(Y) = \frac{21 + 25}{2} = 23 \implies E(X) = 0.1(23) = 2.3$$
$$V(Y) = \frac{5^2 - 1}{12} = 2 \implies V(X) = (0.1)^2(2) = 0.02$$
$$\sigma(X) = \sqrt{0.02} \approx 0.1414$$